

Lab 1 – Measurements and Tolerances

Your Name: _____

Purposes of This Lab

- We will explore the use of four common analog measuring devices, the ruler, the micrometer caliper (mic), the beam balance (scale), and the graduated cylinder.
- We will also explore the use of one common digital measuring device, the digital caliper.
- We will determine the least counts, the ILEs or tolerances for a single measurement, and the zero errors (if any), for all five measuring devices.
- We will learn how to state single measurements for some of the devices.
- We will explore what the word precision means.
- We will learn how to visualize the relative sizes of mm, cm and inch.

Notes

We will not explore the statistical analysis of multiple measurements in this lab.

In this lab, all measurements made will be metric.

All length measurements will originally be in millimeters (mm).

The acronym ILE refers to “instrument limiting error”, or tolerance for a single measurement.

Feel free to use the “Measurement Table Least Count ILE” text file provided online as reference for this lab activity.

Step 1:

A) See **figure 1**.

We will use the top left side of the metric side of ruler for this part of Step 1.

What is the least count of this part of the ruler? Answer: _____ mm

What is the ILE of this part of the ruler? Answer: _____ mm

What fraction of the ruler's least count did you use to determine its ILE?
Answer: _____

What numbers can you write for the least significant digit for a measurement using the ruler? (For example, 0 through 9, the numbers 0, 2, 4, 6 and 8, or the numbers 0 and 5.) Answer: _____

B) See **figure 2**.

We will use the digital caliper for this part of Step 1.

What is the least count of the digital caliper? Answer: _____ mm

What is the ILE of the digital caliper? Answer: _____ mm

What fraction of the digital caliper's least count did you use to determine its ILE? Answer: _____

What numbers can you write for the least significant digit for a measurement using the digital caliper? Answer: _____

C) See **figure 3**.

We will use the micrometer caliper (mic) for this part of Step 1.

What is the least count of the mic? Answer: _____ mm

What is the ILE of the mic? Answer: _____ mm

What fraction of the mic's least count did you use to determine its ILE? Answer: _____

What numbers can you write for the least significant digit for a measurement using the mic? Answer: _____

Step 2: What zero errors, if any, positive or negative, do the above devices have? For the mic, be sure to use the clutch release whenever closing its anvils to check for zero error, or when making a measurement.

Zero errors: ruler: _____ mm

mic: _____ mm digital caliper: _____ mm

Which of these three devices is the most precise? Answer: _____

Step 3: The provided rectangular block's dimensions, in cm, are **given** -- to three significant figures -- as 4.46 cm x 4.46 cm x 1.78 cm, (Length x Width x Depth, or L x W x D).

A) Knowing that an inch is defined to be exactly 2.54 cm, what are the block's dimensions (L x W x D) in inches?

[At this point, it is not necessary to write any tolerances with your answers, but state your answers, in inches, to three significant figures.]

_____ in x _____ in x _____ in (L x W x D)

Now measure the block's dimensions, **in inches**, using the **digital caliper**, rounding your answers, in the normal fashion, to three significant figures.

[See **figure 4**. You will probably have to change the digital caliper's display to show your measurements in inches.]

_____ in x _____ in x _____ in (L x W x D)

Are the measured values for length, width and depth, in inches, the same as the nominal values given above, or are they different?

Answer:

B) Using the **ruler**, now measure the block's width, **in millimeters**. State your answer to three significant figures, but now with the tolerance (ILE) for one measurement using a ruler. Be careful that the least significant digit you write for your best guess and tolerance reflects the ILE of the ruler.

Answer: _____ mm

Did you use the edge of the ruler to make this measurement? Answer: _____

Did you notice on the ruler that the number 1 refers to 1 cm, not 1 mm?

[One millimeter is about as thick as 10 sheets of ordinary paper. One centimeter is 10 times as thick, and is about 2/5ths (40%) of an inch.]

C) Measure the block's width again, but this time align one side of the block with the 10 mm or 1 cm mark. **Try to avoid any parallax (or eyeball) error**. As before, state your answer to three significant figures and with the tolerance (ILE) for one measurement using a ruler. Be careful that the least significant digit you write for your best guess and tolerance reflects the ILE of the ruler.

Answer: _____ mm

Comparing the two measurements for the block's width, did you get the same answer, or did you get different answers? Answer: _____

Step 4:

A) You should know how to use a beam balance.

When using the beam balance, be sure to check for, and correct, any zero error.

[Your teacher will show you how to do this.]

What is the least count of the beam balance? Answer: _____ grams (gm)

What is the ILE of the beam balance? Answer: _____ gm

What fraction of the beam balance's least count did you use to determine its ILE?
Answer: _____

What numbers can you write for the least significant digit for a mass measurement using the beam balance? Answer: _____

B) Now measure the mass of the spherical ball provided. Express your answer as your actual measurement (best guess for a single observation), plus or minus the balance beam's ILE. You can use a piece of masking tape to hold the ball in place.

State your answer to three significant figures and with the tolerance (ILE) for one measurement using the beam balance. Be careful that the least significant digit you write for your best guess and tolerance reflects the ILE of the beam balance.

[Example: 42.7 ± 0.1 gm]

Mass M of Spherical Ball: _____ gm

Step 5:

A) See **figure 5**.

In this step we will measure the volume of the ball using a graduated cylinder. We will make measurements, in milliliters (ml) of water level, using a graduated cylinder, **before and after** the insertion of the ball into the graduated cylinder. The difference between these measurements will be the volume of the ball.

When writing down your liquid level measurements, be sure to state your answers to three significant figures and with the tolerance (ILE) for one measurement using the graduated cylinder. Be careful that the least significant digit you write for your best guess and tolerance reflects the ILE of the graduated cylinder.

[Example: 51.5 ± 0.5 ml]

Liquid Level Before Insertion = _____ ml

Liquid Level After Insertion = _____ ml

B) Now take the difference between the "After" and "Before" liquid measurements, retaining as many significant digits as the individual "After" and "Before" measurements. **This the volume of your spherical ball.**

Volume V of the ball = _____ cm³

[Notice that the unit has changed from ml to cm³, because 1 cm³ equals 1 ml.]

Step 6:

Now calculate the **density** (rho, ρ) of the ball by plugging the **Mass M** (from Step 4) and the **Volume V** (from step 5) into this formula:

$$\rho = M/V$$

Answer = _____ gm/cm³

Density of Ball

[The density of the ball should be stated to three significant figures, since both the Mass M and the Volume V were measured to three significant figures. We will not attempt to calculate the “propagated” tolerance or ILE for the ball’s density.]

This completes the “Measurements and Tolerances” lab activity.

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FIGURES



figure 1



figure 2



figure 3



figure 4

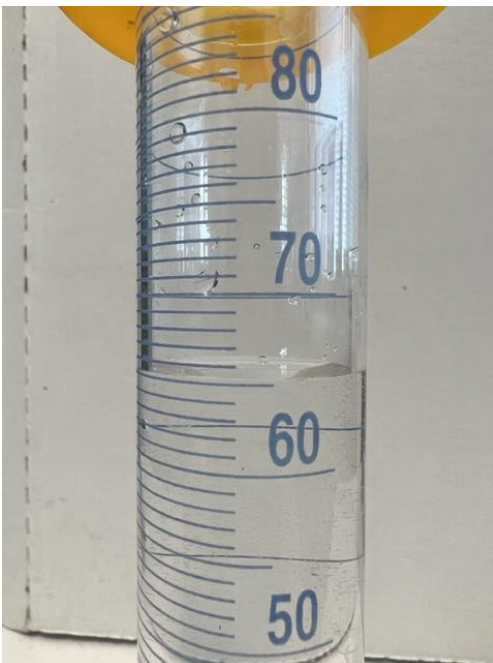


figure 5